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AutoEncoders and VAE

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Autoencoder

- A type of neural network that is trained to learn a compressed representation of input data and generate output matching the input
- Unsupervised learning

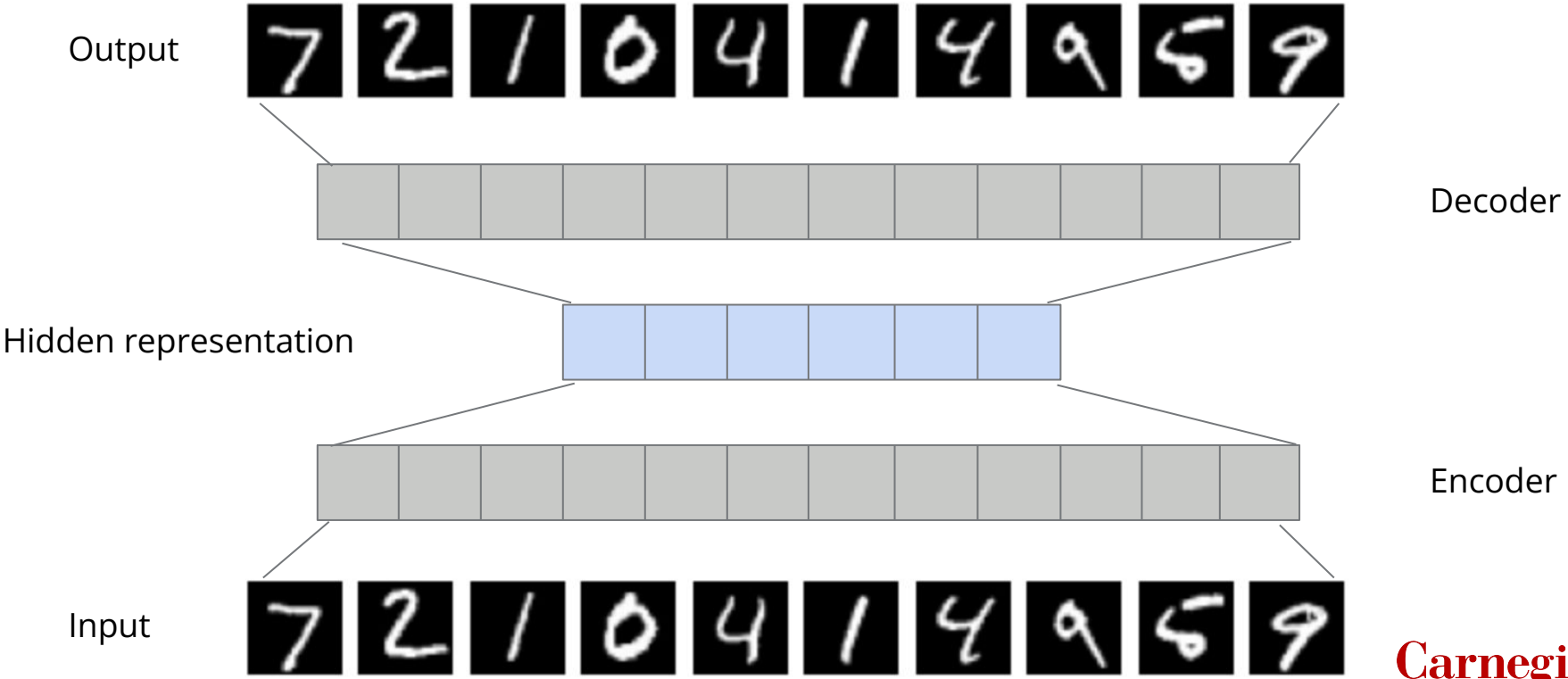
Encoder: Takes in the input data and maps it to a lower-dimensional latent space representation

Decoder: Takes the encoded representation and generates a reconstructed output that closely matches the original input

Why Would I Use An Auto Encoder?

- Feature Extraction
  - Classification
  - Clustering
  - Especially useful when labeling is scarce
- Denoising
  - Audio Data
  - Medical Imaging
  - Sensor Data
- Anomaly Detection
  - Wait a minute this isn't normal
- Dimensionality Reduction
  - Better than PCA for nonlinear Manifolds
- Not great at generative tasks
  - Learning the structure

Autoencoder



Implementation and training

- A sequence of linear or convolutional layers
- Decoder is often a mirror of Encoder

Training objective: Minimize the difference between inputs and outputs
Loss function: MSE Loss, L1 Loss, Binary CrossEntropy Loss (for [0,1] data)

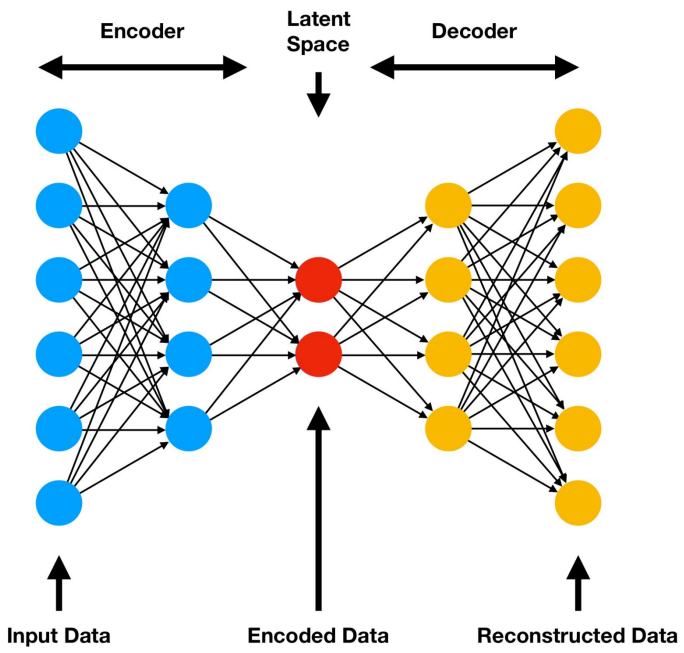
Drawbacks of the AE

Limitations of the Auto Encoder :

1. Limited control over the latent space representation
2. Overfitting
3. Generation not Possible

For a good recent read

https://www.sciencedirect.com/science/article/pii/S0010465525002309



Variational Autoencoders

An Auto Encoder but better

Aim : Learn the data distribution of the input data

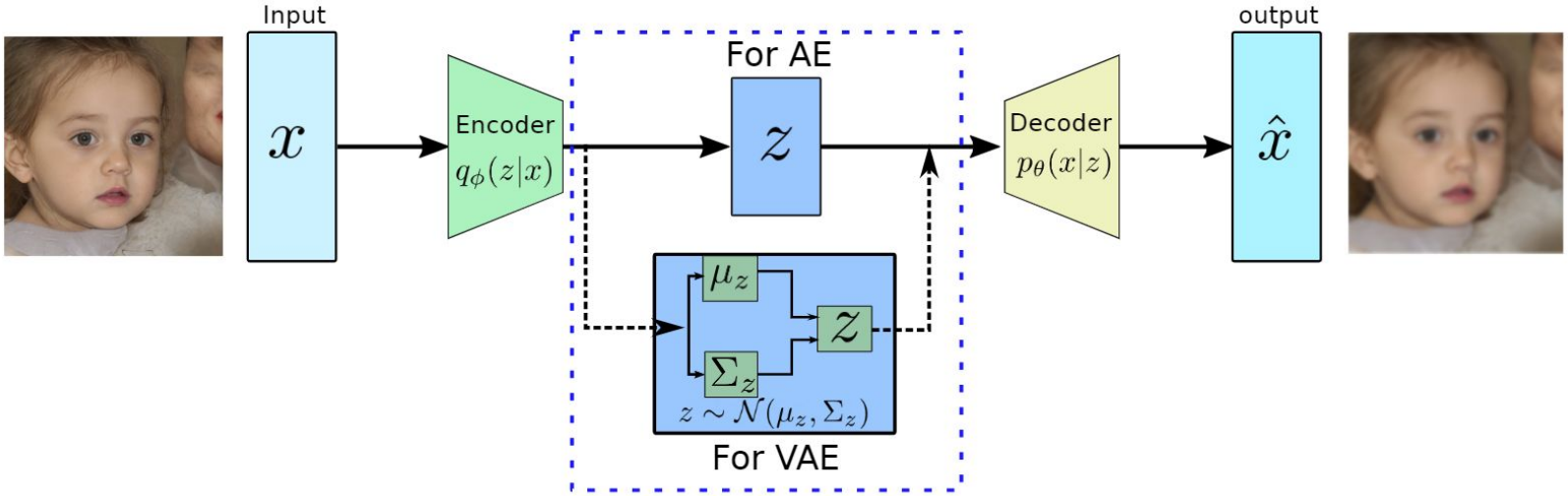
Why?

1. Data and Image Generation
2. Data Compression
3. Anomaly Detection

Why Would I Use A Variational Auto Encoder?

- Generative Modeling
  - Sample New Data from learned distribution
  - Data Augmentation
- Latent Space Exploration
  - Understand Underlying factors of Data
- Uncertainty Awareness
  - Models Distribution
  - Captures Variability

Variational Autoencoders



Variational Autoencoders

- VAEs are probabilistic models to learn data distribution
- Map inputs to a probability distribution
- Objective is to maximize the evidence lower bound (ELBO) : NLL of Data
- Allows learning of a structured latent space representation

Variational Autoencoders

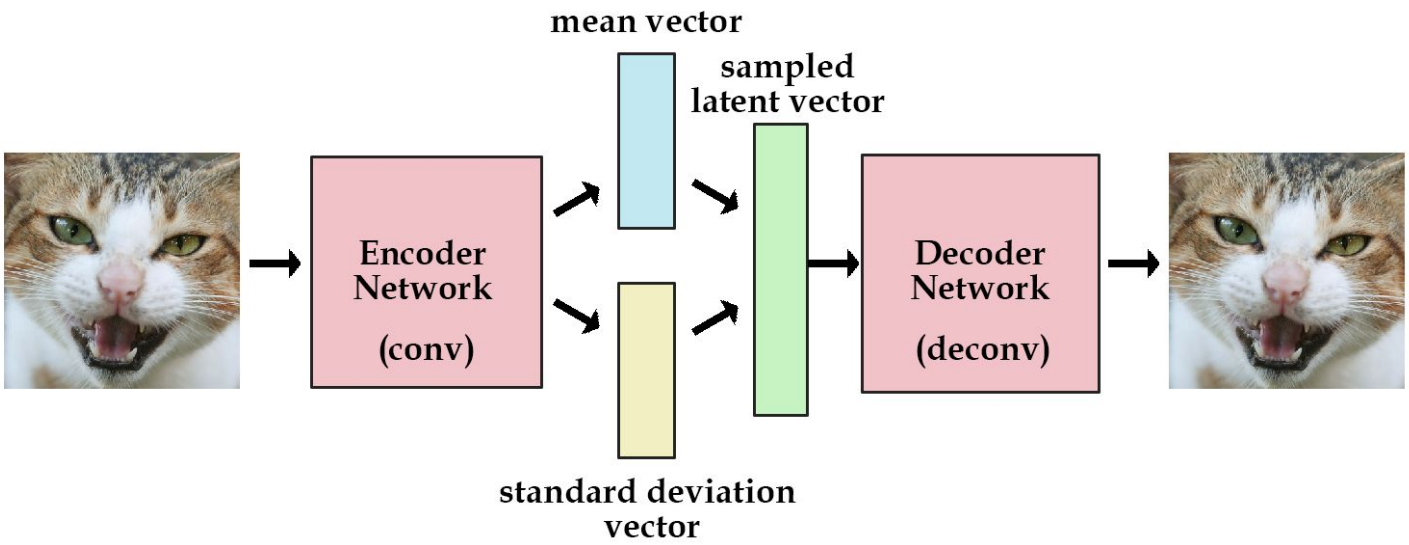
Loss Function

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}^{(i)}) = -D_{KL}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})||p_{\boldsymbol{\theta}}(\mathbf{z})) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}|\mathbf{z})\right]$$

First Term : Forcing latent vector distribution to be Normal(0, 1)

Second Term : Reconstruction Loss

Variational Autoencoders

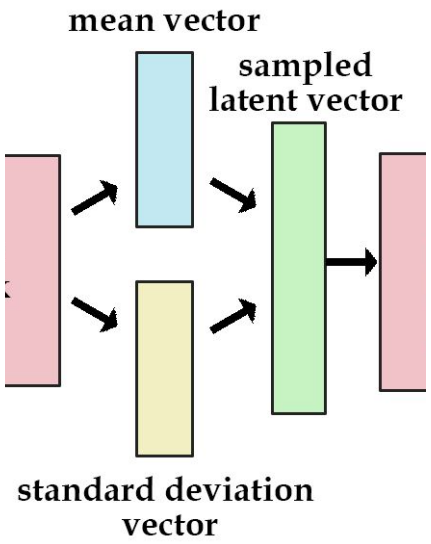


Variational Autoencoders

Re-parametrization Trick

Sampled Vector : Mean Vector + Std deviation vector x Noise

Sampled Vector : $\boldsymbol{\mu} + \boldsymbol{\sigma} \times \boldsymbol{\epsilon}$

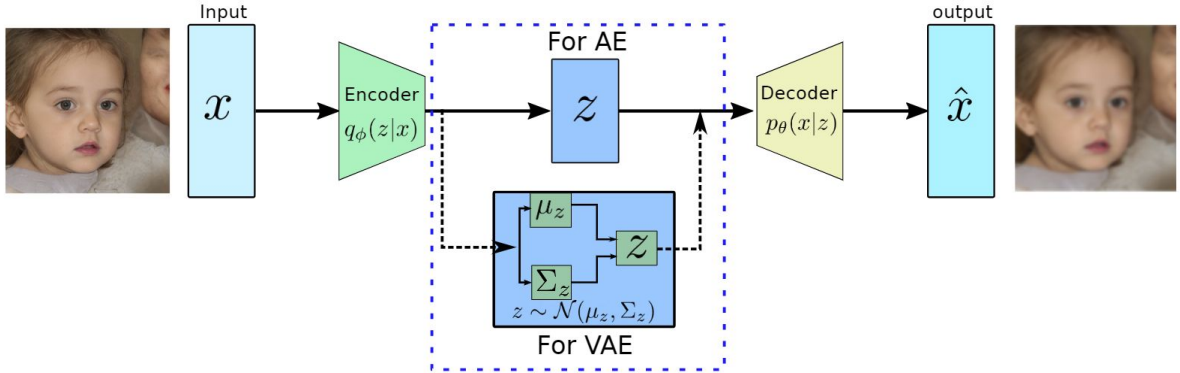


Variational Autoencoders

Issues with AE :

1. Latent Space is unstructured and difficult to interpret

SOLUTION : Map it to a Normal distribution

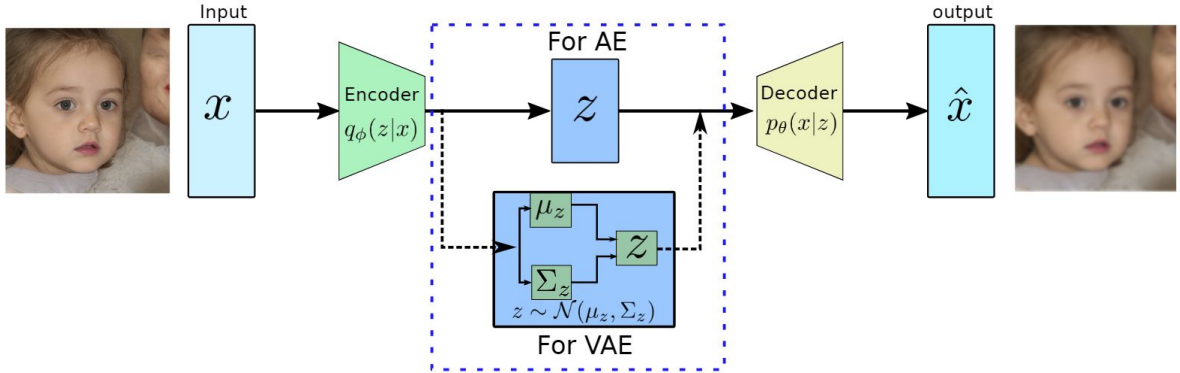


Variational Autoencoders

Issues with AE :

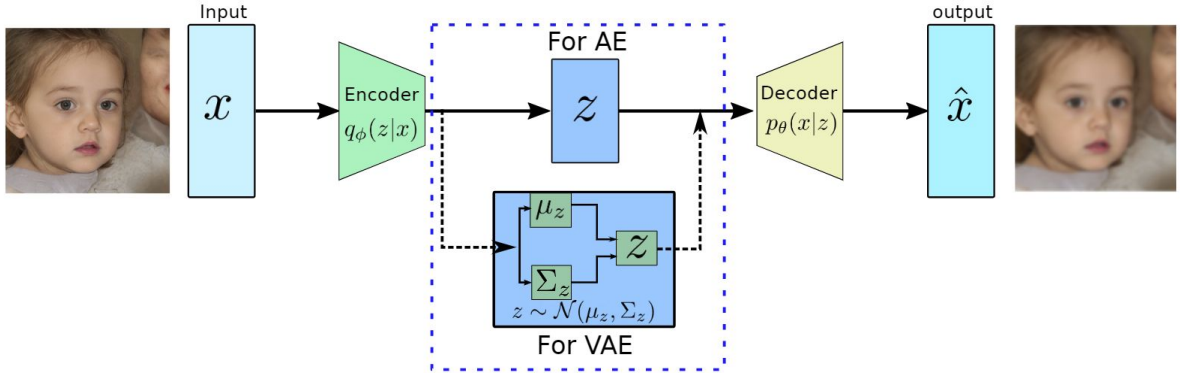
2. Overfitting

SOLUTION : Normal Distribution mapping restricts freedom



Variational Autoencoders

Issues with AE :

3. Data Generation?

SOLUTION : Give a normal sampled vector as an input to the decoder